📐 Geometry Handbook — Formulas & Figures (SSC CGL)

> Every shape & theorem with a **labelled figure** + its formulas. Companion to

> SSC-CGL-ASO-Formulas-Tricks-Master.md.

> Syllabus: Quant — Geometry & Mensuration (Para 13.10.3 / 13.11.1.4–13.11.1.5).

> View in a monospace font (or the HTML/PDF) so the figures stay aligned.

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1. Lines & Angles

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Types of angles: Angle pairs on a transversal:

• Acute < 90° a\ b

• Right = 90° ┐ \

• Obtuse 90–180° ———\———— line 1

• Straight = 180° —— c \ d

• Reflex 180–360° \

e\ f \

————\————\—— line 2

g\ h

`

Concept	Rule
Complementary angles	sum = **90°**
Supplementary angles	sum = **180°**
Angles on a straight line	sum = **180°** (linear pair)
Angles around a point	sum = **360°**
Vertically opposite angles	**equal** (a = d, b = c)
Parallel lines: corresponding	**equal**
Parallel lines: alternate (interior/exterior)	**equal**
Parallel lines: co-interior (allied)	sum = **180°**

2. Triangle — basics & types

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A

/\ • Angle sum ∠A + ∠B + ∠C = 180°

/ \ • Exterior angle = sum of 2 remote interiors

/ \ ∠ACD = ∠A + ∠B

/ \ • Sum of any two sides > third side

/________ D • Side opposite the bigger angle is longer

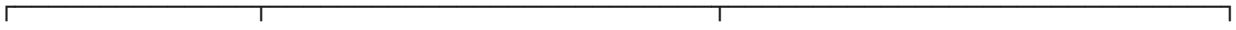
B C

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By sides	By angles
Equilateral — all sides equal (each angle 60°)	**Acute** — all angles < 90°
Isosceles — two sides equal (base angles equal)	**Right** — one angle = 90°
Scalene — all sides different	**Obtuse** — one angle > 90°

3. The Four Centres of a Triangle

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| Centre | Where lines meet | Note |

|---|---|---|

| Centroid G | the 3 MEDIANS | divides each median 2:1 |

| | (vertex → midpoint) | (from vertex) |

| Incentre I | the 3 ANGLE BISECTORS | centre of inscribed circle |

| Circumcentre | the 3 PERP. BISECTORS | centre of circumscribed ○; |

| O | of the sides | equidistant from vertices |

| Orthocentre | the 3 ALTITUDES | — |

| H | (vertex ⊥ opposite side) | |

• G, O, H are collinear → the EULER LINE; and OG : GH = 1 : 2.

• In an equilateral triangle all four centres COINCIDE.

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A Median: A → midpoint of BC.

/|\ Centroid G lies on all three, 2:1 from vertex.

/ | \ r (inradius) = Area / s

/ G \ R (circumradius) = (a·b·c) / (4·Area)

/ /|\ \ In a right triangle: R = hypotenuse / 2

B__/_|___C and the circumcentre is the midpoint of hypotenuse.

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4. Triangle Area Formulas

Given	Area
base & height	**½ × base × height**
3 sides a,b,c (Heron)	**√[s(s−a)(s−b)(s−c)]**, s = (a+b+c)/2
two sides & included angle	**½ · a · b · sin C**
equilateral side a	**(√3 / 4) · a^2**
with inradius r	**r × s**
with circumradius R	**(a·b·c) / (4R)**

`

Equilateral triangle (side a):

/\ height h = (√3/2)·a

/ \ area = (√3/4)·a^2

/ h \ inradius r = a / (2√3)

/_____\ circumradius R = a / √3 (R = 2r)

a

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5. Pythagoras Theorem & Triples

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|\ $c^2 = a^2 + b^2$

| \ (hypotenuse^2 = sum of squares of legs)

b | \ c Common Pythagorean triples:

| \ 3, 4, 5 5, 12, 13

|___\ 8, 15, 17 7, 24, 25

a 9, 40, 41 20, 21, 29

(and any multiple: 6-8-10, 9-12-15 ...)

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Result	Formula
Hypotenuse	$\sqrt{(a^2 + b^2)}$
A leg	$\sqrt{(c^2 - \text{other leg}^2)}$
Altitude to hypotenuse	(a·b)/c
Right-triangle inradius	r = (a + b − c)/2

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6. Similarity & Congruence

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SIMILAR (same shape, sizes scaled) CONGRUENT (identical)

A P Tests: SSS, SAS, ASA, AAS, RHS

/\ /\ (RHS only for right triangles)

/ \ / \ Similar tests: AA, SSS~, SAS~

/___\ /_____\

ratio of sides equal: AB/PQ = BC/QR = CA/RP = k

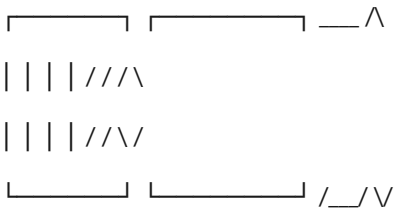
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In similar triangles (ratio k)	Relationship
Corresponding sides	in ratio **k**
Perimeters	in ratio **k**
Areas	in ratio **k^2**
(for solids) Volumes	in ratio **k^3**
Basic Proportionality (Thales)	a line ∥ one side divides the other two **proportionally**

7. Quadrilaterals

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Square Rectangle Parallelogram Rhombus



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Shape	Area	Perimeter / notes
Square (side a)	a^2	4a; diagonal = $a\sqrt{2}$
Rectangle (l, b)	l·b	2(l+b); diagonal = $\sqrt{(l^2+b^2)}$
Parallelogram	base × height	opposite sides ∥ & equal; diagonals bisect
Rhombus (diagonals d_1,d_2)	**$\frac{1}{2} \cdot d_1 \cdot d_2$**	all sides equal; diagonals ⊥ bisect
Trapezium (∥ sides a,b; height h)	**$\frac{1}{2}(a+b) \cdot h$**	one pair of sides parallel
Kite (diagonals d_1,d_2)	$\frac{1}{2} \cdot d_1 \cdot d_2$	diagonals ⊥
Cyclic quad (Brahmagupta)	$\sqrt{[(s-a)(s-b)(s-c)(s-d)]}$	opposite angles sum 180°

8. Polygons

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Regular hexagon (n = 6):

___ • Sum of interior angles = (n − 2) × 180°

/ \ • Each interior angle (regular) = (n−2)·180° / n

/ \ • Each exterior angle (regular) = 360° / n

\ / • Number of diagonals = n(n − 3) / 2

___/

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n	Name	Interior-angle sum	Each interior (regular)
3	Triangle	180°	60°
4	Quadrilateral	360°	90°
5	Pentagon	540°	108°
6	Hexagon	720°	120°
8	Octagon	1080°	135°

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9. Circle — parts & theorems

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_____

/ arc \

/ ___ \ • Radius r; Diameter d = 2r

| / \ ●———|——— radius • Circumference = 2πr = πd

| | C |chord | • Area = πr^2

| ___/ | • Chord: a line joining two points on ○

\ tangent / • Tangent: touches ○ at one point ⊥ radius

\ ___|_____/ • Secant: cuts ○ at two points

|

tangent line

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Theorem	Statement
Angle at centre	= **2 ×** angle at circumference (same arc)
Angles in same segment	**equal**
Angle in a semicircle	**90°** (diameter subtends right angle)
Cyclic quadrilateral	opposite angles sum **180°**
Tangent ⊥ radius	at the point of contact (90°)
Two tangents from a point	are **equal** in length
Alternate segment	tangent-chord angle = angle in alternate segment
Intersecting chords	PA · PB = PC · PD
Equal chords	equidistant from the centre

Circle measure	Formula
Area	πr^2
Circumference	$2 \pi r$
Arc length (angle θ°)	$(\theta/360) \times 2\pi r$
Sector area (angle θ°)	$(\theta/360) \times \pi r^2$
Segment area	sector area − triangle area

9A. Circle — Chord Theorems

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PERPENDICULAR FROM CENTRE BISECTS A CHORD

● O (centre)

/|\

/ | \ r • OM ⊥ AB ⇒ AM = MB

/ | \ • Distance of chord from centre:

/ |M \ OM = $\sqrt{(r^2 - (AB/2)^2)}$

A •————┴————• B • Conversely, the line from O to the

chord AB midpoint of a chord is ⊥ to it.

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Chord theorem	Statement
Perpendicular from centre	**bisects** the chord (AM = MB)
Line from centre to chord-midpoint	is **perpendicular** to the chord
Equal chords	are **equidistant** from the centre (and vice-versa)
Longer chord	is **nearer** the centre; the diameter is the longest chord
Equal chords subtend	**equal angles** at the centre

9B. Intersecting Chords, Secants & Tangent–Secant (Power of a Point) ⭐

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(1) CHORDS meet INSIDE (2) SECANTS meet OUTSIDE (3) TANGENT + SECANT

A P P

\ D \ \

\ / \● A ● T (tangent)

P ● X P ●—\——● B (secant PAB) P ●—\——● A —— ● B

/ \ \ \ (secant PAB)

/ \ \● C

C B \● D (secant PCD)

PA·PB = PC·PD PA·PB = PC·PD PT^2 = PA·PB

(two chords AB, CD (two secants from P) (tangent^2 = secant product)

crossing at X: AX·XB = CX·XD)

`

Theorem (Power of a Point)	Relation
Two chords intersect **inside** at X	AX · XB = CX · XD
Two secants from an **external** point P	PA · PB = PC · PD
Tangent–Secant from external point P	PT^2 = PA · PB
Two **tangents** from external point P	PT_1 = PT_2 (equal lengths)

 One idea: for any point P and a line through it cutting the circle at two points,

the **product of the two distances is constant** (the "power" of P). Inside → chord·chord;

outside → secant·secant; tangent is the limiting case where the two points merge → PT^2.

9C. Tangent Theorems

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TANGENT ⊥ RADIUS TWO TANGENTS FROM A POINT

—— ——

/ \ / \

| ●O | | ●O |

\ | / \ /|\ /

\ | / ————● P (external)

\ |/ 90° PT_1 = PT_2 ; OP bisects ∠T_1PT_2

———●——— tangent ; OT_1 ⊥ PT_1 , OT_2 ⊥ PT_2

T

`

Tangent theorem	Statement
Tangent ⊥ radius	at the point of contact (90°)
Two tangents from external point	are **equal** ($PT_1 = PT_2$)
Line from centre to external point	**bisects** the angle between the two tangents
Tangent length	$PT = \sqrt{(OP^2 - r^2)}$
Alternate Segment Theorem	angle between tangent & chord = angle in the **alternate segment**

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DIRECT & TRANSVERSE COMMON TANGENTS (two circles, centres d apart, radii R, r)

• Length of DIRECT common tangent = $\sqrt{(d^2 - (R - r)^2)}$

• Length of TRANSVERSE common tangent = $\sqrt{(d^2 - (R + r)^2)}$

`

Number of common tangents by position of two circles (centres `d` apart, radii `R ≥ r`):

Condition	Position of Circles	Common Tangents
`d > R + r`	Separate (do not touch)	4
`d = R + r`	Externally touching	3 ⭐
`R − r < d < R + r`	Intersecting	2
`d = R − r`	Internally touching	1 ⭐
`d < R − r`	One circle inside another	0
`d = 0, R = r`	Coincident (same circle)	Infinite

🔑 *Trick:* Tangent count drops as the circles get closer — the two **touching** cases (⭐) are the exam favourites: externally touching → **3**, internally touching → **1**.

9D. Proportionality & Angle Theorems (Circle)

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ANGLE AT CENTRE = 2 × ANGLE AT CIRCUMFERENCE ANGLE IN SAME SEGMENT

O C D

/|\ ∠AOB = 2·∠ACB \ / ∠ACB = ∠ADB

/ | \ \ / (subtend the

A ● | ● B (same arc AB) A ●——X——● B same arc AB)

● C

`

Angle / proportion theorem	Statement
Angle at centre	= **2 ×** angle at the circumference on the same arc
Angles in the same segment	are **equal**
Angle in a semicircle	= **90°**
Cyclic quadrilateral	opposite angles are **supplementary** (sum 180°)
Exterior angle of cyclic quad	= interior **opposite** angle
Ptolemy's theorem (cyclic quad)	AC · BD = AB · CD + AD · BC (product of diagonals)
Equal arcs	subtend equal angles at the centre & equal chords

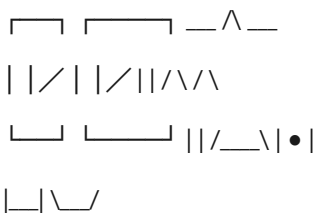
10. Mensuration 2D — areas & perimeters

Shape	Area	Perimeter
Square (a)	a^2	4a
Rectangle (l,b)	l·b	2(l+b)
Triangle (b,h)	½·b·h	a+b+c
Equilateral (a)	$(\sqrt{3}/4)a^2$	3a
Parallelogram	b·h	2(a+b)
Rhombus (d_1,d_2)	½·d_1·d_2	4·side
Trapezium (a,b,h)	½(a+b)h	sum of sides
Circle (r)	πr^2	$2\pi r$
Semicircle (r)	$\frac{1}{2}\pi r^2$	$\pi r + 2r$
Ring (R,r)	$\pi(R^2-r^2)$	—

11. Mensuration 3D — solids

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Cube Cuboid Cylinder Cone Sphere



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Solid	Volume	Curved/Lateral SA	Total SA
Cube (a)	a^3	$4a^2$	$6a^2$; diagonal $a\sqrt{3}$
Cuboid (l,b,h)	l·b·h	2h(l+b)	2(lb+bh+hl); diag $\sqrt{(l^2+b^2+h^2)}$
Cylinder (r,h)	$\pi r^2 h$	$2\pi rh$	$2\pi r(r+h)$
Cone (r,h)	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(r+l)$; slant $l=\sqrt{(r^2+h^2)}$
Sphere (r)	$4/3\cdot\pi r^3$	—	$4\pi r^2$
Hemisphere (r)	$2/3\cdot\pi r^3$	$2\pi r^2$	$3\pi r^2$
Frustum (R,r,h)	$\frac{1}{3}\pi h(R^2+r^2+Rr)$	$\pi(R+r)l$	$+ \pi(R^2+r^2)$

`

Cone slant height: Sphere: Hemisphere:

/| 4 2 ___

/ | h $l^2 = r^2+h^2$ $V = \frac{4}{3}\pi r^3$ $V = \frac{2}{3}\pi r^3$ / \

l / | 3 3 |____|

/___| SA = $4\pi r^2$ TSA = $3\pi r^2$ (flat + curved)

r

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12. Coordinate Geometry

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y

| • $P(x_1,y_1)$, $Q(x_2,y_2)$

Q •| _____

| •P Distance = $\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$

———┼——— x Midpoint = $((x_1+x_2)/2 , (y_1+y_2)/2)$

| Slope m = $(y_2-y_1)/(x_2-x_1)$

| Section (m:n) = $((mx_2+nx_1)/(m+n), (my_2+ny_1)/(m+n))$

`

Concept	Formula
Distance	$\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$
Midpoint	$((x_1+x_2)/2, (y_1+y_2)/2)$
Slope m	$(y_2-y_1)/(x_2-x_1)$
Line (slope-intercept)	$y = mx + c$
Parallel lines	equal slopes ($m_1 = m_2$)
Perpendicular lines	$m_1 \cdot m_2 = -1$
Area of triangle (vertices)	½ \	$x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\$	

13. Quick recall table

Prompt	Answer
Triangle angle sum	180°
Equilateral area	$(\sqrt{3}/4)\cdot a^2$
Heron's area	$\sqrt{[s(s-a)(s-b)(s-c)]}$
Centroid divides median	2 : 1
Circumradius R	abc/(4·Area)
Inradius r	Area/s
Angle in semicircle	90°
Angle at centre vs circumference	2 : 1
Rhombus area	$½\cdot d_1\cdot d_2$
Trapezium area	$½(a+b)\cdot h$
Interior-angle sum (n-gon)	$(n-2)\cdot 180°$
Diagonals of n-gon	$n(n-3)/2$
Cube diagonal	$a\sqrt{3}$
Cuboid diagonal	$\sqrt{(l^2+b^2+h^2)}$
Cone slant l	$\sqrt{(r^2+h^2)}$
Sphere volume / SA	$4/3\cdot\pi r^3$ / $4\pi r^2$
Hemisphere TSA	$3\pi r^2$
Areas of similar figures	ratio of sides^2
Perpendicular slopes	$m_1\cdot m_2 = -1$

✅ **Strategy:** learn the **figure first**, then attach its formulas — recall is far

stronger when you can *picture* the shape. Redraw 2–3 figures from memory daily.

Back to the main formula sheet ».